

ICT-2107: Signals and Systems

CLASS 1 — STUDENT LECTURE NOTE

Course Overview and Introduction to Signals and Systems

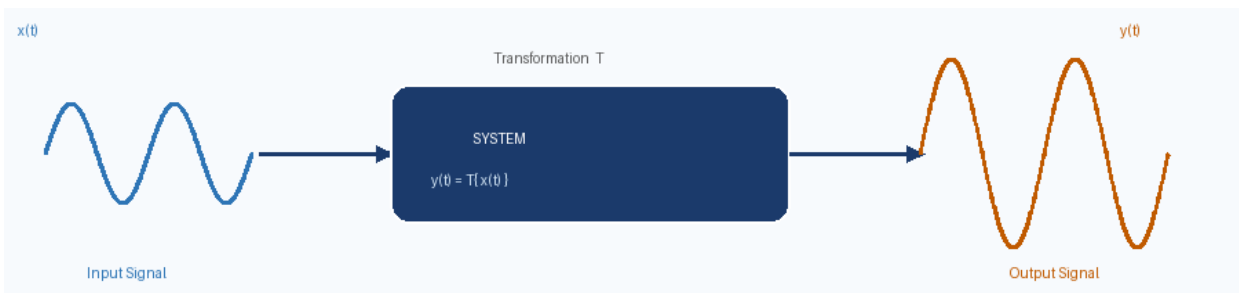


Figure 1. A system receives an input signal $x(t)$, transforms it, and produces an output signal $y(t)$.

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Class Duration	Target Students	Course Type
1 Hour	B.Sc. Engineering — 2nd Year	Theory

Part 1 — What is a Signal?

1.1 Definition of a Signal

A signal is a function of one or more independent variables that carries information about the behaviour or nature of some physical phenomenon. Think of it this way: whenever a physical quantity — pressure, voltage, temperature, light intensity, position — changes with time (or space), that variation is a signal. The variation is not random noise; it is structured information about what is happening in the real world.

Formal definition

A signal x is a mathematical function such that its value $x(t)$ — or $x[n]$ in the discrete case — at each value of the independent variable t (or n) represents a measurable physical quantity of interest.

The independent variable is most often time t , but it can also be spatial position (in images), frequency, depth, or any other measurable dimension.

The simplest way to understand a signal is through an everyday example. When you speak, your lips push and pull the air around them. This causes air pressure at every nearby point to rise and fall rhythmically. That rhythmic variation in pressure — as a function of time — is your speech signal. It carries the complete information about the words you said, the tone you used, and even your identity as a speaker.

1.2 Mathematical Notation — The Most Important Convention

Two types of signals are used throughout this course. They differ in the nature of the independent variable. The notation difference is critical — never confuse them:

Continuous-Time Signal	Written as $x(t)$ — round brackets. The variable t is a real number and can take any value in an interval, including fractions like 0.001 s or irrational numbers like $\sqrt{2}$ s. The signal is defined at every instant of time with no gaps.
Discrete-Time Signal	Written as $x[n]$ — square brackets. The variable n is an integer: ..., -2, -1, 0, 1, 2, 3, ... The signal has a value only at these isolated instants. Between integers the signal simply does not exist.

Why notation matters

If you write $x(t)$ and mean $x[n]$, your equations will be wrong. The differential equations used for continuous-time systems (Laplace transform) are completely different from the difference equations used for discrete-time systems (z-transform). Get the bracket right from day one, and every later topic will follow naturally.

1.3 Continuous-Time Signals — Deep Explanation with Figures

A continuous-time signal is an unbroken curve when plotted against time. No matter how much you zoom in, there is always a signal value — the curve never has a jump unless the physical phenomenon itself jumps (like a switch turning on).

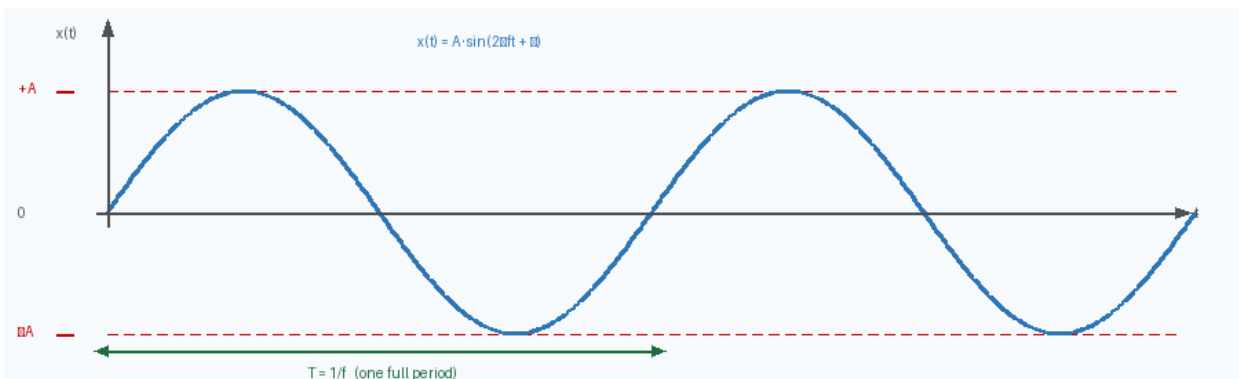


Figure 2. Continuous-time sinusoidal signal $x(t) = A \cdot \sin(2\pi ft)$. Amplitude A , period $T = 1/f$, and the repetitive structure are all annotated.

The sinusoidal signal is the single most important continuous-time signal in this course because, by Fourier analysis, every other signal can be expressed as a sum of sinusoids. Its equation is:

$$x(t) = A \cdot \sin(2\pi ft + \phi)$$

A (Amplitude)	The maximum value the signal reaches. Units depend on the physical quantity — volts (V) for electrical signals, Pascals (Pa) for pressure, etc.
f (Frequency)	Number of complete cycles per second. Unit: Hertz (Hz). If $f = 100$ Hz, the signal completes 100 full oscillations every second.

T (Period)	Time for one complete cycle. $T = 1/f$. If $f = 100$ Hz then $T = 0.01$ s = 10 ms.
ϕ (Phase)	Shifts the signal left or right in time. Unit: radians. If $\phi = 0$, the signal starts at zero and rises.
ω_0 (Angular frequency)	$\omega_0 = 2\pi f$. Unit: radians per second (rad/s). This is the natural unit for Fourier and Laplace analysis.

Other common continuous-time signals you will encounter in this course include:

1. **Real exponential:** $x(t) = A \cdot e^{(\alpha t)}$. If $\alpha < 0$ the signal decays to zero (like a capacitor discharging). If $\alpha > 0$ it grows (like an unstable feedback system).
2. **Complex exponential:** $x(t) = e^{(j\omega_0 t)} = \cos(\omega_0 t) + j \cdot \sin(\omega_0 t)$ by Euler's formula. This is the eigenfunction of LTI systems — you will use it constantly from Lecture 5 onwards.
3. **Unit step:** $u(t) = 1$ for $t \geq 0$, $u(t) = 0$ for $t < 0$. Models a switch being turned on at $t = 0$.
4. **Unit impulse:** $\delta(t)$ — an infinitely tall, infinitely narrow spike at $t = 0$ with unit area. The impulse response of a system to $\delta(t)$ completely characterises the system.

1.4 Discrete-Time Signals — Deep Explanation with Figures

A discrete-time signal $x[n]$ has values only at integer indices. On a plot, it appears as a collection of vertical stems — not as a continuous curve. The gaps between stems are not zero; the signal is simply undefined there. This is the kind of signal a digital system produces or processes.

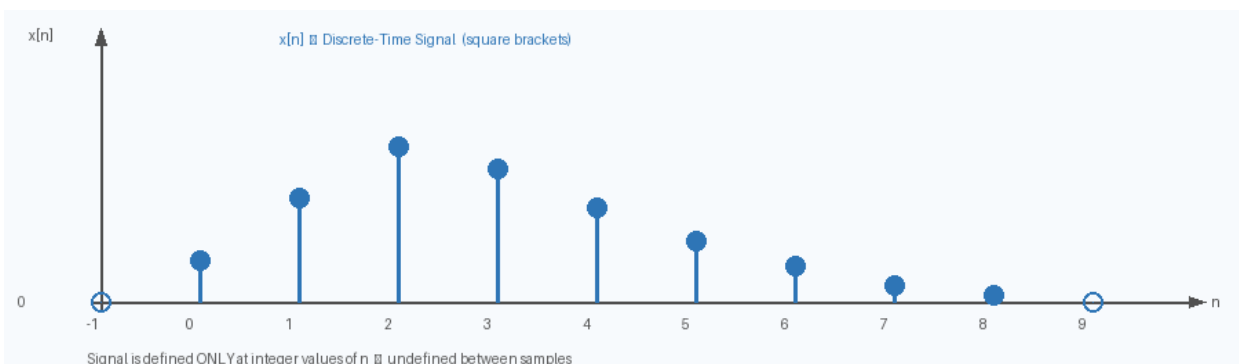


Figure 3. Discrete-time signal $x[n]$ shown as a stem plot. Filled circles mark sample values. The signal is undefined between integer indices.

Discrete-time signals arise naturally when a continuous-time signal is sampled at regular time intervals. If we sample a continuous-time signal $x(t)$ every T_s seconds (the sampling period), we obtain the discrete-time sequence:

$$x[n] = x(n \cdot T_s)$$

where T_s = sampling period (seconds), $f_s = 1/T_s$ = sampling rate (Hz)

A concrete example: a telephone system samples speech at $f_s = 8\,000$ Hz (8 kHz), meaning $T_s = 1/8000 = 0.000125$ s = 125 μ s. Every 125 microseconds, the system takes one snapshot of the speech voltage and stores it as a number. The resulting sequence $x[n] = x(n \times 0.000125)$ is a discrete-time signal.

Sampling theorem (preview)

To perfectly reconstruct a continuous-time signal from its samples, the sampling rate f_s must be at least twice the highest frequency $f_{\max}$ present in the signal:

$f_s \geq 2 \cdot f_{\max}$ (Nyquist–Shannon Sampling Theorem)

For telephone speech with $f_{\max} \approx 4$ kHz, we need $f_s \geq 8$ kHz — exactly the telephone standard. You will study this theorem fully in Section 13 of the course.

1.5 Continuous-Time vs Discrete-Time — Side-by-Side

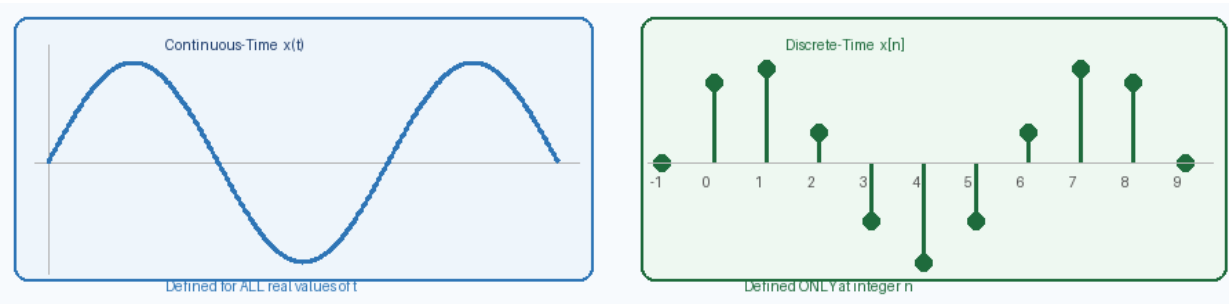


Figure 4. Left: continuous-time signal $x(t)$ — an unbroken smooth curve. Right: discrete-time signal $x[n]$ — values only at integer indices n .

Property	Continuous-Time $x(t)$	Discrete-Time $x[n]$
Independent variable	t — any real number ($-\infty < t < \infty$)	n — integers only ($\dots, -2, -1, 0, 1, 2, \dots$)

Notation	Round brackets $x(t)$	Square brackets $x[n]$
Graph	Smooth or piecewise-smooth curve	Stem plot (vertical lines + filled circles)
Defined between samples?	Yes — at every instant	No — undefined between integers
Mathematical tool	Differential equations, Laplace transform	Difference equations, z-transform
Fourier tool	Fourier Transform (CTFT)	Discrete-Time Fourier Transform (DTFT)
Typical origin	Physical sensor, transducer	ADC output, digital computer, sensor with clock
Examples	ECG, AC voltage, speech	Digital audio, pixel values, stock prices

1.6 A Signal is a Mathematical Function — Worked Example

Understanding that a signal is a function — a rule that assigns one number to each value of the independent variable — is the most important conceptual step in this course. Once you see signals as functions, all the mathematics becomes natural.

Example: $x(t) = 5 \sin(2\pi t)$

This signal has amplitude $A = 5$, frequency $f = 1$ Hz, period $T = 1$ s, and angular frequency $\omega_0 = 2\pi$ rad/s. Let us compute its value at five specific instants and understand what each one means physically:

$x(t) = 5 \sin(2\pi t)$			
$x(0.00) = 5 \sin(0)$	$= 5 \times 0.000$	$= 0.000$	← starts at zero, rising
$x(0.25) = 5 \sin(\pi/2)$	$= 5 \times 1.000$	$= +5.000$	← peak (maximum)
$x(0.50) = 5 \sin(\pi)$	$= 5 \times 0.000$	$= 0.000$	← crosses zero, falling

$$\begin{aligned}
 x(0.75) &= 5 \sin(3\pi/2) &= 5 \times -1.000 &= -5.000 &\leftarrow \text{trough} \\
 &&&&&(\text{minimum}) \\
 x(1.00) &= 5 \sin(2\pi) &= 5 \times 0.000 &= 0.000 &\leftarrow \text{one full} \\
 &&&&&\text{cycle complete}
 \end{aligned}$$

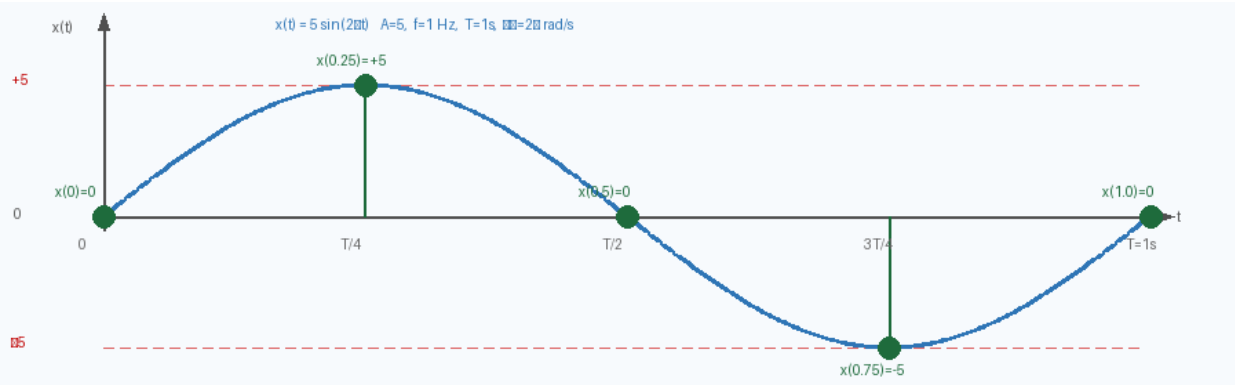


Figure 5. Plot of $x(t) = 5 \sin(2\pi t)$. Green dots mark the five computed sample values. The signal completes exactly one cycle in $T = 1$ second.

The physical interpretation: if this signal represents an AC voltage, the voltage at the terminal rises from 0 V to +5 V in a quarter second, returns to 0 V at half a second, drops to -5 V at three-quarters of a second, and returns to 0 V after one full cycle. An oscilloscope would show exactly this shape.

Generalization to other frequencies

What changes when the frequency changes? Only the rate of oscillation. The shape stays the same:

$$\begin{aligned}
 f &= 50 \text{ Hz} \quad (\text{AC mains in Bangladesh}) : & x(t) &= 220\sqrt{2} \cdot \sin(2\pi \cdot 50 \cdot t) & T &= 20 \text{ ms} \\
 f &= 440 \text{ Hz} \quad (\text{musical note A4}) : & x(t) &= A \cdot \sin(2\pi \cdot 440 \cdot t) & T &= 2.27 \text{ ms} \\
 f &= 2.4 \text{ GHz} \quad (\text{Wi-Fi carrier}) : & x(t) &= A \cdot \sin(2\pi \cdot 2.4 \times 10^9 \cdot t) & T &= 0.417 \text{ ns}
 \end{aligned}$$

All three are sinusoidal signals with the same mathematical form — only the frequency parameter f changes. This universality is exactly why the sinusoid is so powerful as an analytical tool.

1.7 Real-Life Examples of Signals

Signal	Independent Variable	Physical Quantity	Information Carried
Speech	Time t (seconds)	Air pressure (Pa)	Words, tone, speaker identity, emotion
ECG (electrocardiogram)	Time t (seconds)	Voltage (mV) at chest	Heart rate, arrhythmia, cardiac health
Temperature	Time t (hours)	Temperature ($^{\circ}\text{C}$)	Thermal variation, HVAC control signal
Image (photograph)	Space: x, y (pixels)	Light intensity (0–255 per channel)	Scene content, colour, faces, text
Video	Time t + space x, y	Intensity sequence	Moving visual information (30–120 fps)
Internet data	Time t (seconds)	Voltage/optical power	Binary bits: 0 and 1
Radio/TV broadcast	Time t (seconds)	Electromagnetic field amplitude	Audio, video, data payload
EEG (brain)	Time t (seconds)	Voltage (μV) at scalp	Brain activity, sleep stage, seizure
GPS navigation	Time t (seconds)	RF carrier phase	Position (latitude, longitude, altitude)
Seismic signal	Time t (seconds)	Ground displacement (μm)	Earthquake, underground structure



Figure 6. Speech-like signal. The amplitude varies irregularly with time, encoding the words spoken. Voiced segments have large amplitude; silences have near-zero amplitude.

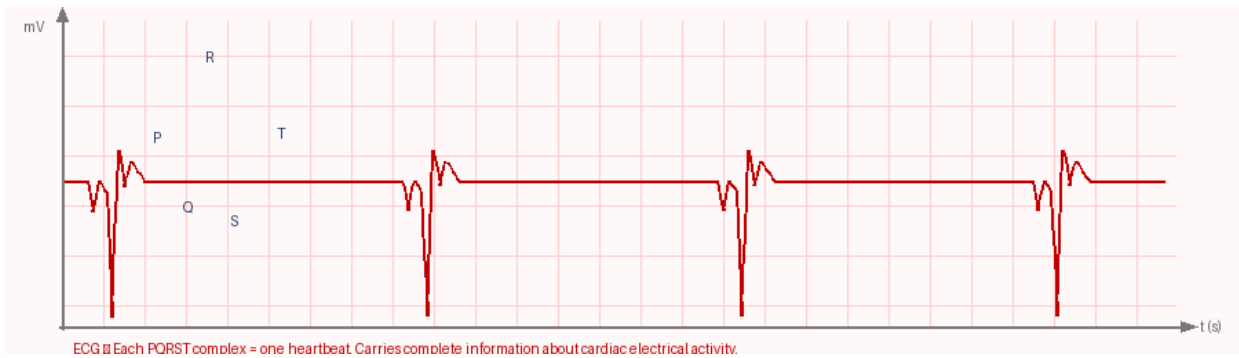


Figure 7. ECG signal. Each PQRST complex represents one heartbeat. The P-wave is atrial depolarisation; the tall R-spike is ventricular contraction; the T-wave is ventricular repolarisation.

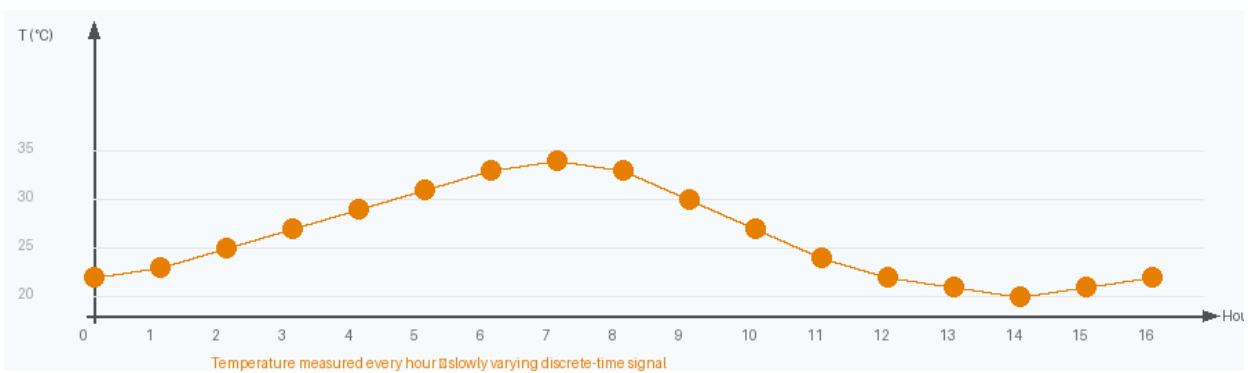


Figure 8. Temperature signal recorded hourly. This is a slowly varying discrete-time signal. The peak around noon and the trough around midnight are clearly visible.

Part 2 — What is a System?

2.1 Definition of a System

A system is any physical device, process, algorithm, or mathematical model that takes one or more input signals, performs an operation on them, and produces one or more output signals. The system does not create information — it transforms signals from one form or quality to another.

Formal definition

A system is characterised by its transformation T , which maps an input signal x to an output signal y :

$$y(t) = T\{x(t)\} \quad (\text{continuous-time system})$$

$$y[n] = T\{x[n]\} \quad (\text{discrete-time system})$$

The box diagram $x(t) \rightarrow [T] \rightarrow y(t)$ is the standard representation used throughout this course and in every engineering textbook.

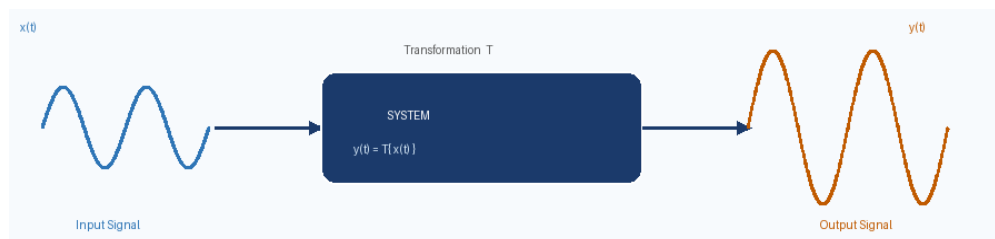


Figure 9. Block diagram of a general system. The box represents the transformation T . The input $x(t)$ enters from the left; the output $y(t)$ exits to the right.

The power of this block-diagram abstraction is that you do not need to know the internal physics of a system in order to analyse it mathematically. A microphone, a digital filter running on a DSP chip, and a human cochlea all perform different physical processes — but each one can be represented as the same simple box transforming an input signal into an output signal.

2.2 The Input-Output Model — Detailed Explanation

The input-output model is the foundation on which the entire course is built. Once we know the transformation T that a system performs, we can:

1. Predict the output for any given input, without building the physical system.
2. Design the system to achieve a desired output for a given input.
3. Analyse system properties: is it linear? Time-invariant? Causal? Stable? Memoryless?
4. Cascade multiple systems by composing their transformations.

For most practical engineering systems, the transformation T can be described by a differential equation (continuous-time) or a difference equation (discrete-time). A key goal of this course is to give you the mathematical tools to solve these equations efficiently — without simulation, using Fourier series, Fourier transforms, Laplace transforms, and z-transforms.

2.3 Mathematical System Examples — Step by Step

Example 1: Amplifier — $y(t) = 2 \cdot x(t)$

The simplest non-trivial system multiplies the input by a constant. This models an ideal linear amplifier with gain 2:

$$y(t) = 2 \cdot x(t)$$

Input: $x(t) = 3 \sin(2\pi \cdot 100 \cdot t)$

Output: $y(t) = 2 \times 3 \sin(2\pi \cdot 100 \cdot t) = 6 \sin(2\pi \cdot 100 \cdot t)$

Effect: amplitude doubled ($3 \rightarrow 6$), frequency unchanged (100 Hz), shape unchanged

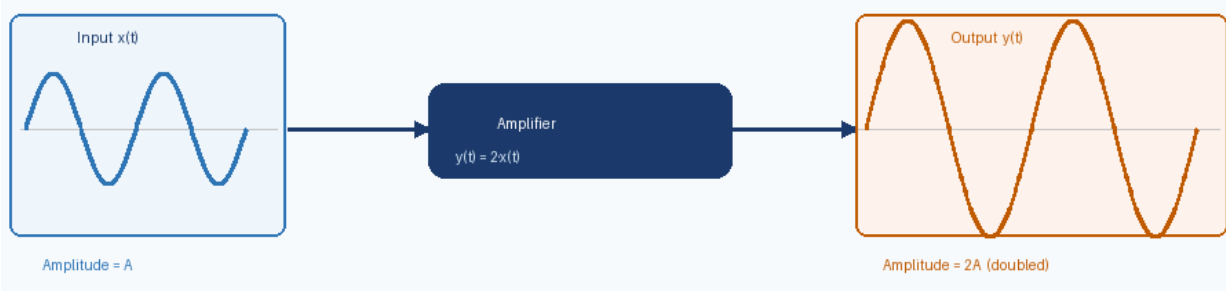


Figure 10. Amplifier system $y(t) = 2x(t)$. The output (right, orange) has exactly twice the amplitude of the input (left, blue) but the same frequency and waveform shape.

Notice: only the amplitude changed. The frequency, phase, and waveform shape are all preserved. This is a property of linear systems — they cannot create new frequencies, only scale existing ones. You will prove this mathematically when we study linearity.

Example 2: Time Delay — $y(t) = x(t - t_0)$

A delay system shifts the input signal in time by t_0 seconds:

$$y(t) = x(t - t_0)$$

If $x(t) = \sin(2\pi t)$ and $t_0 = 0.25$ s:

$$y(t) = \sin(2\pi(t - 0.25)) = \sin(2\pi t - \pi/2) = -\cos(2\pi t)$$

Effect: same frequency, same amplitude, but the waveform is delayed by $t_0 = T/4$

Physical example: if you shout in a canyon, the echo you hear is the same signal delayed by $2d/c$ seconds, where d is the distance to the canyon wall and $c = 340$ m/s is the speed of sound. The delay system models the propagation of a signal through a medium.

Example 3: Integrator — $y(t) = \int x(\tau) d\tau$

An integrating system accumulates the input signal over time. It is the continuous-time analog of a running sum:

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

If $x(t) = A$ (a constant DC signal):

$$y(t) = \int_0^t A d\tau = A \cdot t \quad (\text{output grows linearly with time})$$

If $x(t) = A \cdot \cos(2\pi f t)$:

$$y(t) = A / (2\pi f) \cdot \sin(2\pi f t) \quad (\text{output is a sinusoid with reduced amplitude})$$

Physical example: a capacitor is an integrating system. The voltage across a capacitor $v_C(t)$ is the integral of the current flowing into it: $v_C(t) = (1/C) \int i(\tau) d\tau$. This is why capacitors are used as integrators in analog circuits.

Example 4: Differentiator — $y(t) = dx(t)/dt$

A differentiating system computes the instantaneous rate of change of the input:

$$y(t) = d/dt [x(t)] = x'(t)$$

If $x(t) = A \cdot \sin(2\pi f t)$:

$$y(t) = A \cdot 2\pi f \cdot \cos(2\pi f t)$$

Effect: same frequency, amplitude scaled by $2\pi f$, phase shifted by $+\pi/2$ (90°)

The amplitude of the output is $2\pi f$ times larger than the input. This means a differentiator amplifies high frequencies much more than low frequencies — the opposite of an integrator. A physical inductor is a differentiating element: $v_L(t) = L \cdot di(t)/dt$.

Example 5: Moving Average (Discrete-Time) — $y[n] = (x[n] + x[n-1])/2$

In the discrete-time world, the moving average is a fundamental signal processing operation. It reduces rapid fluctuations (noise) while preserving slow trends:

$$y[n] = (x[n] + x[n-1]) / 2$$

Given: $x[n] = \{4, 8, 2, 6, 0, 10\}$ for $n = 0, 1, 2, 3, 4, 5$

$$y[0] = (x[0] + x[-1]) / 2 = (4 + 0) / 2 = 2.0 \quad (\text{assume } x[-1]=0)$$

$$y[1] = (x[1] + x[0]) / 2 = (8 + 4) / 2 = 6.0$$

$$y[2] = (x[2] + x[1]) / 2 = (2 + 8) / 2 = 5.0$$

$$y[3] = (x[3] + x[2]) / 2 = (6 + 2) / 2 = 4.0$$

$$y[4] = (x[4] + x[3]) / 2 = (0 + 6) / 2 = 3.0$$

$$y[5] = (x[5] + x[4]) / 2 = (10 + 0) / 2 = 5.0$$

Compare $x[n] = \{4, 8, 2, 6, 0, 10\}$ with $y[n] = \{2, 6, 5, 4, 3, 5\}$. The rapid jump from 8 to 2 in x is smoothed to 6 to 5 in y . The moving average is a simple low-pass filter — it reduces high-frequency variations while preserving the overall trend. This is the principle behind smoothing financial data, sensor readings, and audio signals.

2.4 Microphone as a System — Complete Analysis

A microphone is one of the most important transducer systems in ICT engineering. Understanding it in complete detail builds the intuition for every other system in this course.



Figure 11. Microphone system analysis. The sound pressure wave $p(t)$ is the input; the electrical voltage $v(t)$ is the output. The transformation is electromechanical transduction.

Microphone system equation

Input signal: $p(t)$ — air pressure variation (Pa) at the microphone diaphragm.

Output signal: $v(t)$ — electrical voltage (mV or μ V) proportional to the pressure.

System equation: $v(t) = S \cdot p(t)$

where S = microphone sensitivity, typically 1–50 mV/Pa for common condenser microphones.

Example: if $S = 10 \text{ mV/Pa}$ and the speaker produces $p(t) = 2 \cdot \sin(2\pi \cdot 440 \cdot t) \text{ Pa}$ (a 440 Hz tone),

then $v(t) = 10 \times 2 \cdot \sin(2\pi \cdot 440 \cdot t) = 20 \cdot \sin(2\pi \cdot 440 \cdot t)$ mV.

The output signal has the same frequency (440 Hz) and the same waveform shape as the input. Only the physical form (pressure $\rightarrow$ voltage) and the scale (mV/Pa) have changed.

Inside the microphone, the physical mechanism depends on the type. In a condenser (capacitor) microphone, sound pressure moves a thin diaphragm, changing the capacitance between the diaphragm and a fixed backplate. This changing capacitance is converted to a changing voltage. In a dynamic microphone, the diaphragm is attached to a coil that moves inside a magnetic field, generating a voltage by electromagnetic induction. Despite these different mechanisms, both are modelled identically at the signal level: $v(t) = S \cdot p(t)$.

2.5 Filter System — Detailed Analysis

A filter is a system that selectively removes certain frequency components from a signal while preserving others. Filters are among the most widely used systems in all of engineering.

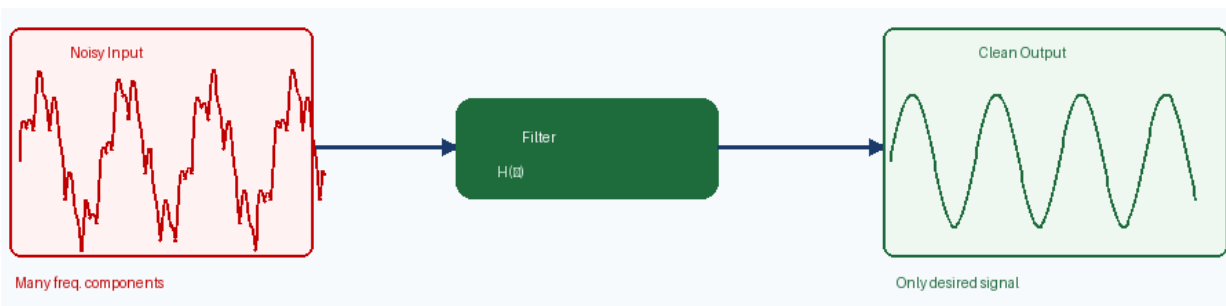


Figure 12. Filter system $H(\omega)$. The noisy input (left, red) contains unwanted high-frequency components. After filtering, the clean output (right, green) retains only the desired signal.

Consider a telephone speech signal corrupted by high-frequency electrical noise:

$$x(t) = 3 \cdot \sin(2\pi \cdot 300 \cdot t) + 0.8 \cdot \sin(2\pi \cdot 4000 \cdot t) + 0.4 \cdot \sin(2\pi \cdot 8000 \cdot t)$$

↑ desired speech ↑ high-freq noise ↑ more noise

A low-pass filter with cut-off frequency $f_c = 400$ Hz passes frequencies below 400 Hz and blocks frequencies above 400 Hz. Its output is approximately:

$$y(t) \approx 3 \cdot \sin(2\pi \cdot 300 \cdot t)$$

The noise components at 4 kHz and 8 kHz are eliminated.
Speech quality improves significantly.

Filter Type	Passes	Blocks	Application
Low-pass (LPF)	Frequencies below f_c	Frequencies above f_c	Noise removal, anti-aliasing, audio tone control
High-pass (HPF)	Frequencies above f_c	Frequencies below f_c	Removing DC offset, treble boost, edge detection
Band-pass (BPF)	Frequencies between f_L and f_H	All others	Radio tuner, equaliser band, telephone band
Band-stop (BSF)	All except band f_L – f_H	Specific band	Removing 50/60 Hz power line hum from audio

You will design and analyse all four types of filters in Section 13 of this course using the Laplace transform and frequency response methods developed in earlier sections.

2.6 Real-Life Examples of Systems

System	Input Signal	Operation Performed	Output Signal
Microphone	Sound pressure $p(t)$	Electromechanical transduction: $v(t) = S \cdot p(t)$	Electrical voltage $v(t)$
Loudspeaker	Electrical voltage $v(t)$	Electroacoustic transduction	Sound pressure $p(t)$
Camera (lens+sensor)	Light intensity $I(x,y)$	Photoelectric conversion + A/D sampling	Digital image array $x[m,n]$
Amplifier	Weak signal $x(t)$	Multiplication: $y(t) = G \cdot x(t)$	Amplified signal $y(t)$

Low-pass filter	Noisy signal	Remove frequencies above f_c	Clean signal
ADC	Continuous voltage $x(t)$	Sampling + quantisation	Digital sequence $x[n]$
DAC	Digital sequence $x[n]$	Reconstruction + interpolation	Analog voltage $x(t)$
Equaliser (audio)	Music signal	Boost/cut frequency bands	Adjusted music signal
JPEG encoder	Raw image $x[m,n]$	DCT + quantisation + entropy coding	Compressed image file
Human cochlea (ear)	Sound wave $p(t)$	Spectral decomposition by basilar membrane	Nerve impulses by frequency

Part 3 — Mobile Communication: A Complete System Chain

Mobile communication is the richest and most immediately relatable example of signals and systems in everyday life. Every phone call, text message, and video stream you send or receive passes through a chain of many systems, each transforming the signal in a specific way. Understanding this chain at a conceptual level will give you a purpose for every topic in this course.

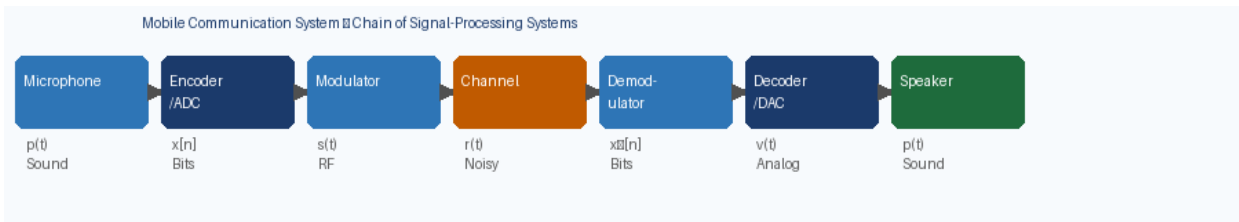


Figure 13. Mobile communication system chain. Each block is a system. The signal is transformed at every stage, from sound wave to radio wave and back to sound wave.

3.1 Stage-by-Stage Analysis

Let us trace the complete journey of a voice signal from speaker's mouth to listener's ear:

Stage 1: Microphone (Transducer System)

Input: $p(t)$ — sound pressure wave from your voice. Frequency range: 80 Hz to 8 kHz for speech.

Operation: $v(t) = S \cdot p(t)$ — electromechanical transduction (sensitivity $S \approx 10 \text{ mV/Pa}$).

Output: $v(t)$ — continuous-time analog electrical voltage signal.

Course connection: This is a continuous-time LTI system (Sections 2–3 of the course).

Stage 2: Encoder / Analog-to-Digital Converter (Sampling System)

Input: $v(t)$ — continuous-time analog voltage.

Operation: Sample at $f_s = 8000 \text{ Hz}$ (telephone) or 44100 Hz (CD audio).

$$x[n] = v(n \cdot T_s) \text{ where } T_s = 1/f_s.$$

Each sample is then quantised to 8 bits (telephone) or 16 bits (CD) — 256 or 65536 levels.

Output: $x[n]$ — discrete-time digital sequence (a stream of binary numbers).

This is the boundary between the analog and digital worlds.
Course connection: Sampling theorem, Section 13 of this course.

Stage 3: Modulator (Signal-Transformation System)

Input: $x[n]$ — digital bit sequence representing the speech.
Operation: Map bits onto a high-frequency radio carrier wave using modulation.
In 4G/5G: OFDM (Orthogonal Frequency Division Multiplexing).
 $s(t) = \text{Re}\{ \sum_k X[k] \cdot e^{j2\pi k \Delta f \cdot t} \}$ — a sum of complex exponentials at different frequencies.
This is a direct application of the Discrete Fourier Transform (DFT).
Output: $s(t)$ — bandpass RF (radio frequency) signal ready for transmission.
Course connection: Fourier Transform (Sections 6–7), complex exponentials.

Stage 4: Wireless Channel (System with Distortion and Noise)

Input: $s(t)$ — transmitted radio signal.
Operation: The channel introduces three kinds of distortion:
(a) Path loss: signal power decreases as $1/d^2$ with distance d .
(b) Multipath: signal arrives via multiple reflected paths with different delays.
 $r(t) = \sum_i a_i \cdot s(t - \tau_i)$ — a sum of delayed, attenuated copies.
(c) Additive noise: $r(t) = s(t) * h(t) + n(t)$
where $h(t)$ = channel impulse response, $n(t)$ = noise.
Output: $r(t)$ — received signal at the other phone, distorted by channel and noise.
Course connection: Convolution, impulse response, LTI systems (Sections 2–3).

Stage 5: Demodulator (Inverse-Transformation System)

Input: $r(t)$ — received (distorted) radio signal.
Operation: Reverse the modulation to recover the digital bit sequence.
In OFDM: compute the DFT of the received signal to separate frequency subcarriers.
Equalization corrects channel distortion: $\hat{X}[k] = R[k] / H[k]$ (frequency-domain division).
Output: $\hat{x}[n]$ — estimated digital bit sequence (ideally = $x[n]$).
Course connection: Fourier Transform, convolution theorem, frequency response.

Stage 6: Decoder / Digital-to-Analog Converter (Reconstruction System)

Input: $\hat{x}[n]$ — digital sequence.

Operation: DAC converts each sample back to a voltage level.

$\hat{v}(t) = \sum_n \hat{x}[n] \cdot \text{sinc}(t/T_s - n)$ — ideal interpolation (Shannon reconstruction formula).

In practice, a low-pass reconstruction filter replaces the sinc interpolation.

Output: $\hat{v}(t)$ — reconstructed analog voltage signal, approximately equal to the original $v(t)$.

Course connection: Sampling theorem, reconstruction filter (Section 13).

Stage 7: Loudspeaker (Transducer System)

Input: $\hat{v}(t)$ — reconstructed electrical voltage signal.

Operation: Electroacoustic transduction — the reverse of the microphone.

$\hat{p}(t) \approx (1/S) \cdot \hat{v}(t)$ — the speaker converts voltage to sound pressure.

Output: $\hat{p}(t)$ — sound wave at the listener's ear. The listener hears the original voice.

Course connection: LTI system analysis, transducer modelling.

3.2 Why This Matters — Connecting Every Stage to a Course Topic

Mobile System Stage	Signal-Processing Operation	Course Section
Microphone / Speaker	Transducer — LTI system modelling	Sections 2–3
ADC (Encoder)	Sampling — $x[n] = x(nT_s)$	Section 13
Modulator (OFDM)	Discrete Fourier Transform	Sections 6–7
Wireless Channel	Convolution: $r(t) = s(t)*h(t) + n(t)$	Sections 2–3
Demodulator (OFDM)	IDFT, frequency-domain equalisation	Sections 6–7
DAC (Decoder)	Signal reconstruction, low-pass filter	Section 13
Audio filter / equaliser	Filter design using frequency response	Sections 9, 13
Echo canceller	Adaptive filter (LTI system)	Sections 2–3, 9

Speech codec (compression)	Fourier representation, predictive coding	Sections 5–7
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The big picture

Every mathematical tool you will study in ICT-2107 — the impulse response, the convolution integral, the Fourier series, the Fourier transform, the Laplace transform, the z-transform, filter design — is directly used somewhere in the mobile communication chain shown above. When you feel lost in an abstract derivation, come back to this chain and ask: "Which stage does this formula describe?"

Part 4 — Why Signals and Systems Matters for ICT Students

The following table connects each area of ICT engineering directly to the signals-and-systems concepts it depends on. As you read through this, notice how the same small set of tools — convolution, Fourier transform, sampling — appears in nearly every application.

ICT Application Area	Signals involved	Systems / Operations	Concepts from this course
Mobile / wireless communication	Speech, RF waves, video	Modulator, channel, demodulator, equaliser	Fourier Transform, convolution, sampling, LTI systems
Audio processing (music, speech)	Microphone signal, music	Equaliser, compressor, reverb, noise canceller	Fourier series, filter design, z-transform
Digital image processing	Photograph pixel array	Sharpening, blurring, edge detection, compression	2D Fourier Transform, convolution
Biomedical engineering	ECG, EEG, EMG, blood pressure	Heart rate detector, arrhythmia classifier	Fourier analysis, filter design, LTI systems
Internet of Things (IoT)	Temperature, pressure, humidity	Sensor signal conditioning, ADC, data transmission	Sampling, A/D conversion, LTI filtering
Control systems	Actuator position, motor speed	PID controller, servo drive	Laplace transform, frequency response, stability
Radar and sonar	Transmitted pulse, received echo	Pulse compressor, range-Doppler processor	Convolution, matched filter, Fourier Transform
Digital TV and video streaming	Video frames, audio	Compression (H.264/H.265), error correction	DCT (Fourier-based), filter banks, sampling

Machine learning / AI	Feature vectors from audio/image	Feature extraction front-end	Fourier features, filter banks, signal statistics
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Bottom line

Signals and Systems is not a theoretical exercise. Every engineering task you will encounter as an ICT graduate — whether in communication, biomedical devices, audio, imaging, control, or AI — requires you to describe a signal mathematically, characterise the system that processes it, and predict or design the output. That is exactly what this course teaches.

Part 5 — Important Terms and Definitions

Every term below was introduced in today's class. Memorise the definitions precisely — engineering exams and subsequent courses require exact language.

Term	Precise Definition	Example / Notes
Signal	A function of one or more independent variables that conveys information about a physical phenomenon.	$x(t)=5\sin(2\pi t)$, ECG waveform, image $f(x,y)$
Continuous-time signal	Signal defined for every real value of t . Notation: $x(t)$ with round brackets.	AC voltage, speech, ECG
Discrete-time signal	Signal defined only at integer values of n . Notation: $x[n]$ with square brackets.	Digital audio samples, pixel values
System	Device, process, algorithm, or model that transforms input signals into output signals.	Amplifier, filter, ADC, microphone
Input signal	The signal applied to the input of a system. Also called the excitation.	Sound wave applied to a microphone
Output signal	The signal produced at the output of a system. Also called the response.	Voltage from a microphone
Transformation T	The mathematical operation of a system: $y(t)=T\{x(t)\}$.	$T\{x(t)\}=2x(t)$ for an amplifier of gain 2
Amplitude A	The maximum absolute value a signal reaches.	$A=5$ in $x(t)=5\sin(2\pi t)$
Frequency f	Number of complete cycles per second. Unit: Hz. $f=1/T$.	440 Hz for musical note A4
Period T	Time for one complete cycle. $T=1/f$.	$T=1/440 \approx 2.27$ ms for 440 Hz
Angular frequency ω_0	$\omega_0=2\pi f$. Unit: rad/s. Natural unit in Fourier and Laplace analysis.	$\omega_0=2\pi \times 50 = 314.2$ rad/s for 50 Hz mains

Phase ϕ	Time shift of a sinusoidal signal. Unit: radians.	$\phi=\pi/2$ shifts sin to cos
Signal processing	Any operation on a signal to modify, analyse, or extract information from it.	Filtering, compression, modulation
Transducer	System that converts a signal from one physical domain to another.	Microphone (sound→voltage), speaker (voltage→sound)
Sampling	Taking values of $x(t)$ at discrete instants: $x[n]=x(nT_s)$.	CD audio: $T_s=1/44100$ s
Sampling rate f_s	Number of samples taken per second. $f_s=1/T_s$. Unit: Hz.	Telephone: 8 kHz, CD: 44.1 kHz
Low-pass filter	System that passes frequencies below cut-off f_c and blocks higher frequencies.	Removes noise from speech signal

Part 6 — In-Class Discussion Questions with Full Answers

Work through these questions with your group during class. For each question, write your answer in the box provided in your notebook before reading the model answer below. A good engineer always justifies their answer using the precise definitions from Part 5.

Q1 Is human voice a signal? Explain precisely why.

Model Answer

Yes. Human voice is a continuous-time signal $p(t)$ where p represents the air pressure at the listener's location and t represents time. The value of $p(t)$ changes rapidly and in a complex pattern as the speaker's vocal cords vibrate and the vocal tract shapes the sound. This variation carries information — the words spoken, the speaker's identity, emotional tone, and accent. Since it is a function of time that conveys information about a physical phenomenon (vocal production), it satisfies the definition of a signal exactly. Notation: $p(t)$ Pa, typical frequency range 80 Hz – 8 kHz.

Q2 Is a mobile phone a system? Identify at least four sub-systems inside it.

Model Answer

Yes, a mobile phone is a complex system — in fact, it is a chain of many cascaded sub-systems. The key sub-systems include: (1) Microphone: input = sound $p(t)$, output = voltage $v(t)$. (2) ADC (Analog-to-Digital Converter): input = voltage $v(t)$, output = digital samples $x[n]$. (3) Audio Codec: input = $x[n]$, output = compressed bit stream. (4) RF Modulator: input = bits, output = radio-frequency signal $s(t)$. (5) Antenna: transmits $s(t)$ as electromagnetic waves. (6) Demodulator, DAC, and Speaker for the receive path. Each sub-system satisfies the definition: it takes an input signal and produces an output signal by performing a transformation T .

Q3 Is a photograph a signal? What are its independent variables?

Model Answer

Yes, a photograph is a 2D signal. It can be written as $f(x, y)$ where x and y are horizontal and vertical spatial coordinates (in pixels or metres), and f is the light intensity (or colour) at that point. At each coordinate pair (x, y) , the signal value represents how bright that point is. A colour photograph has three signal components: $R(x, y)$, $G(x, y)$, and $B(x, y)$ for the red, green, and blue channels respectively. So a 4000×3000 colour photograph is actually three discrete-

time 2D signals, each with 12 million sample values. The independent variables are x and y (spatial position), not time.

Q4 A system is described by $y(t)=3x(t)-2$. Is this a linear system? Why?

Model Answer

No, this system is NOT linear. A system is linear if it satisfies two conditions: (1) Homogeneity: $T\{a \cdot x(t)\} = a \cdot T\{x(t)\}$ for any constant a . Test: $T\{a \cdot x\} = 3(ax) - 2 = 3ax - 2$, but $a \cdot T\{x\} = a(3x - 2) = 3ax - 2a$. These are equal only if $a=1$ — so homogeneity fails. (2) Additivity: $T\{x_1 + x_2\} = T\{x_1\} + T\{x_2\}$. Test: $T\{x_1 + x_2\} = 3(x_1 + x_2) - 2$, but $T\{x_1\} + T\{x_2\} = (3x_1 - 2) + (3x_2 - 2) = 3x_1 + 3x_2 - 4$. These differ by 2, so additivity also fails. The system $y(t)=3x(t)-2$ is an AFFINE (not linear) system because of the constant -2 term. Truly linear systems must satisfy $y=0$ when $x=0$, which this system does not (when $x=0$, $y=-2 \neq 0$).

Q5 What are the input and output of a speaker? Write the system equation.

Model Answer

Input: $v(t)$ — electrical voltage signal delivered by an amplifier (unit: V or mV). Output: $p(t)$ — sound pressure wave produced in the air by the speaker's cone (unit: Pa). System equation: $p(t) \approx (1/S) \cdot v(t)$, where S is the speaker sensitivity in Pa/V. More accurately, the speaker has a frequency-dependent response: $P(\omega) = H(\omega) \cdot V(\omega)$ in the frequency domain, where $H(\omega)$ is the speaker's frequency response function. This means the speaker does not reproduce all frequencies equally — it has a frequency response that rolls off at very low frequencies (< 80 Hz for a typical speaker) and very high frequencies (> 18 kHz for a typical tweeter). You will analyse this using the Fourier transform in Section 9 of the course.

Part 7 — Short Class Quiz

Answer each question in 1–3 sentences or with a clear equation. Use precise definitions and correct notation. Time: 10 minutes.

1. Define a signal. Your definition must include the word "function" and the word "information".
2. Define a system. Your definition must include "input signal", "output signal", and "transformation".
3. Write the general input-output equation of a continuous-time system and draw the block diagram.
4. What is the difference between $x(t)$ and $x[n]$? Why do we use different brackets?
5. A signal is $x(t) = 4 \cos(2\pi \cdot 200 \cdot t)$. Identify: (a) amplitude, (b) frequency, (c) period, (d) angular frequency.
6. Give two examples of signals from biomedical engineering and state what physical quantity each represents.
7. What is the input of a microphone? What is the output? Write the system equation.
8. A system has $y(t) = 2x(t)$. If $x(t) = 3 \sin(2\pi \cdot 50 \cdot t)$, what is $y(t)$? Describe in words what changed.
9. A moving-average system is described by $y[n] = (x[n] + x[n-1])/2$. Compute $y[0], y[1], y[2]$ if $x[n] = \{6, 4, 8\}$ for $n=0, 1, 2$ (assume $x[-1]=0$).
10. Name three sub-systems inside a mobile phone and state the input and output signal of each.

Worked answer to Q5

$$x(t) = 4 \cos(2\pi \cdot 200 \cdot t)$$

$$(a) \text{ Amplitude } A = 4$$

$$(b) \text{ Frequency } f = 200 \text{ Hz}$$

$$(c) \text{ Period } T = 1/f = 1/200 = 0.005 \text{ s} = 5 \text{ ms}$$

$$(d) \text{ Angular frequency } \omega_0 = 2\pi f = 2\pi \times 200 = 1256.6 \text{ rad/s}$$

Worked answer to Q9

$$x[n] = \{6, 4, 8\} \text{ for } n = 0, 1, 2. \text{ Assume } x[-1] = 0.$$

$$y[0] = (x[0] + x[-1]) / 2 = (6 + 0) / 2 = 3.0$$

$$y[1] = (x[1] + x[0]) / 2 = (4 + 6) / 2 = 5.0$$

$$y[2] = (x[2] + x[1]) / 2 = (8 + 4) / 2 = 6.0$$

The system averages consecutive pairs of samples, smoothing rapid changes.

Part 8 — Class Summary and Homework

8.1 Complete Summary

Class 1 — Key Take-Aways

1. A signal is a function $x(t)$ or $x[n]$ that carries information about a physical phenomenon.
 - Continuous-time $x(t)$: defined for all real t . Round brackets.
 - Discrete-time $x[n]$: defined only at integers n . Square brackets.
2. A system performs transformation T : input $x \rightarrow$ output y .
 - Equation: $y(t) = T\{x(t)\}$ Block diagram: $x(t) \rightarrow [T] \rightarrow y(t)$
3. Key signal parameters: Amplitude A , Frequency f (Hz), Period $T=1/f$, Angular frequency $\omega_0=2\pi f$ (rad/s), Phase ϕ (rad).
4. Worked system examples today:
 - Amplifier: $y(t) = 2 \cdot x(t)$ — amplitude doubled
 - Delay: $y(t) = x(t-t_0)$ — signal shifted in time
 - Integrator: $y(t) = \int x(\tau) d\tau$ — accumulates input
 - Differentiator: $y(t) = dx/dt$ — rate of change of input
 - Moving average: $y[n] = (x[n] + x[n-1])/2$ — smooths noise
5. Mobile communication uses 7 systems in a chain:

Microphone $\rightarrow$ Encoder/ADC $\rightarrow$ Modulator $\rightarrow$ Channel $\rightarrow$ Demodulator $\rightarrow$ DAC $\rightarrow$ Speaker.
6. Filter types: Low-pass, High-pass, Band-pass, Band-stop.
7. Every topic in this course corresponds to a real system in engineering.

8.2 Homework Assignment

Submit before the next class. Show all working. Use correct notation throughout.

Part A — Definitions

1. Write precise definitions (in your own words, without looking at notes) for: signal, system, input signal, output signal, continuous-time signal, discrete-time signal, signal processing.

$x(t) = 8 \cos(2\pi \cdot 150 \cdot t + \pi/4)$, identify: (a) amplitude, (b) frequency, (c) period, (d) angular frequency, (e) phase, (f) compute $x(t)$ at $t = 1/600$ s.

2. Draw the stem plot of $x[n] = \{0, 1, 3, 5, 3, 1, 0\}$ for $n = 0, 1, 2, 3, 4, 5, 6$ using square-bracket notation and labelled axes.

Part B — Examples

1. Give five real-life examples of signals not mentioned in the lecture. For each: state the physical quantity, the independent variable, and the information carried.
2. Give five real-life examples of systems not in the lecture. For each: identify the input signal, the output signal, and the mathematical operation performed.
3. Draw input-output block diagrams for: (a) microphone, (b) loudspeaker, (c) digital camera, (d) mobile phone, (e) audio equaliser. Inside each box write the operation T .

Part C — Problems

$y(t) = x(t) + x(t-0.5)$. If $x(t) = 2 \sin(2\pi \cdot 1 \cdot t)$, compute $y(t)$. Plot $x(t)$ and $y(t)$ for $0 \leq t \leq 2$ s on the same axes. Describe in words what the system does.

$y[n] = x[n] - x[n-1]$. Compute $y[0]$ through $y[4]$ for input $x[n] = \{1, 3, 6, 10, 15\}$ ($n=0..4$, $x[-1]=0$). What operation does this system perform? (Hint: compare with mathematical differentiation.)

1. A mobile phone transmits speech sampled at $f_s = 8000$ Hz. (a) What is the sampling period T_s ? (b) How many samples are taken per minute of speech? (c) If each sample is stored as 8 bits, how many bytes are needed for 1 minute of speech?
2. Explain why Signals and Systems is important for ICT students. Give four specific applications from different areas (communication, biomedical, audio, IoT) and state which course topics are relevant to each.

Teacher's Closing Statement

Today we opened ICT-2107: Signals and Systems. We established that a signal is any function that carries information, and a system is any transformation that maps one signal to another. We worked through five mathematical system examples in detail, analysed the microphone and filter as complete systems, and traced a voice call from mouth to ear through a seven-stage communication chain. Every topic in the fourteen weeks ahead — Fourier series, Fourier transform, Laplace transform, z-transform, filter design — is a tool for solving the same fundamental problem: understanding how systems transform signals.

From the next class we will study the classification of signals: continuous-time, discrete-time, analog, digital, periodic, aperiodic, even, and odd. Please read Oppenheim & Willsky Chapter 1, Sections 1.1–1.3 before you arrive.

— End of Class 1 Lecture Note —

